HANKEL DETERMINANTS OF DISCRETE MEASURES WITH POWER-LAW ATOMS, AND OF ZETA VALUES

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ABSTRACT. Let $\sigma = \sum_{k\ge1} w_k \delta_{x_k}$ be a discrete measure whose atoms are regularly spaced with $x_k \sim Ck^{-p}$ for some $p > 0$, and whose weights satisfy $\sum_k w_k < \infty$ and $w_k \ge c_1 k^{-s}$. We prove that the Hankel determinants $D_n = \det(\int x^{i+j}\,\mathrm{d}\sigma)_{i,j=0}^{n-1}$ satisfy

$$\log D_n = -pn^2 \log n + \Big(\tfrac{3p}{2} + \log\big(c(p)C\big)\Big)n^2 + o(n^2), \qquad c(p) = \tfrac{1}{4}\Big(\tfrac{\sqrt{\pi}\,\Gamma(1/p+1)}{\Gamma(1/p+1/2)}\Big)^p.$$

The constant comes from an explicitly solvable constrained equilibrium problem: the maximiser of the logarithmic energy $\iint \log|u^{-p} - v^{-p}|\,\mathrm{d}\mu(u)\,\mathrm{d}\mu(v)$ among probability measures $\mu \le \mathrm{d}u$ on $(0,\infty)$ is unique, it is an average of arcsine laws, and its density is a regularised incomplete beta function. As a corollary, for every real $a > 0$ and real b with $2a + b > 1$, $\log\det_{1\le i,j\le n} \zeta(a(i+j)+b) = -an^2\log n + (\frac{3a}{2} + \log c(a))n^2 + o(n^2)$. For positive integers a, b this gives the leading-order behaviour $-an^2 \log n$ expected by Haynes and Zudilin, together with the n^2 term.

1. INTRODUCTION

1.1. **Setting and main result.** For a positive Borel measure σ on $\mathbb{R}$ with finite moments $m_r = \int x^r\,\mathrm{d}\sigma(x)$, write

$$D_n(\sigma) := \det\big(m_{i+j}\big)_{i,j=0}^{n-1} \qquad (n \ge 1)$$

for its Hankel determinants. We study discrete measures

$$\sigma = \sum_{k\ge1} w_k\,\delta_{x_k}, \qquad x_k = f(k), \tag{1.1}$$

whose atoms accumulate at a single point with polynomial spacing, the model case being $x_k = Ck^{-p}$. Throughout, $p > 0$ and

$$I(p) := \frac{\sqrt{\pi}}{2}\,\frac{\Gamma(\frac{1}{p}+\frac{1}{2})}{\Gamma(\frac{1}{p}+1)}, \qquad c(p) := \frac{1}{4}\Big(\frac{\pi}{2I(p)}\Big)^p, \qquad \kappa_p := \frac{3p}{2} + \log c(p). \tag{1.2}$$

For instance $I(1) = \pi/4$, $c(1) = \frac{1}{2}$, and $I(2) = 1$, $c(2) = \pi^2/16$.

Theorem 1.1. *Let $p > 0$ and $C > 0$. Let $f \in C^1[1,\infty)$ satisfy*

$$\lim_{\xi\to\infty} \xi^{p+1} f'(\xi) = -pC, \qquad 0 < \inf_{\xi\ge1} \xi^{p+1}|f'(\xi)| \le \sup_{\xi\ge1} \xi^{p+1}|f'(\xi)| < \infty, \tag{1.3}$$

and let the weights $w_k \ge 0$ satisfy

$$W := \sum_{k\ge1} w_k < \infty \qquad \text{and} \qquad w_k \ge c_1 k^{-s} \quad (k \ge 1) \tag{1.4}$$

for some constants $c_1 > 0$ and $s \in \mathbb{R}$. Then the measure σ of (1.1) satisfies

$$\log D_n(\sigma) = -pn^2 \log n + \big(\kappa_p + \log C\big)n^2 + o(n^2) \qquad (n \to \infty). \tag{1.5}$$

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Under (1.3), f is strictly decreasing with a finite limit L, and $f(\xi) - L \sim C\xi^{-p}$; Hankel determinants are invariant under translation of σ (Lemma 2.2), so L plays no role. The upper bound in (1.5) uses only $\sum_k w_k < \infty$; apart from the finiteness of the moments, the lower bound uses only $w_k \geq c_1 k^{-s}$.

Since σ has infinitely many atoms, it has monic orthogonal polynomials of all degrees and a Jacobi matrix with off-diagonal entries $b_j > 0$ (Lemma 2.4). Theorem 1.1 is equivalent to a doubly averaged statement about these coefficients.

Corollary 1.2. *Under the hypotheses of Theorem 1.1,*

$$\frac{2}{n^2}\sum_{j=1}^{n-1}(n-j)\log\bigl(j^p b_j\bigr) \longrightarrow \log\bigl(c(p)C\bigr).$$

The constant κ_p is the value of a variational problem, which we solve explicitly. Let

$$\mathcal{A} := \bigl\{\mu \text{ positive Borel measure on } (0,\infty) :\ \mu\bigl((0,\infty)\bigr) = 1,\ \mu \leq \mathrm{Leb}\bigr\},$$

where $\mu \leq \mathrm{Leb}$ means $\mu(B) \leq |B|$ for every Borel set B, and for $\mu \in \mathcal{A}$ put

$$\mathcal{J}_p(\mu) := \iint \log\bigl|u^{-p} - v^{-p}\bigr|\,\mathrm{d}\mu(u)\,\mathrm{d}\mu(v) \in [-\infty,\infty). \tag{1.6}$$

For $\beta > 0$ let ω_β be the arcsine probability measure on $[0,\beta]$, with density $1/(\pi\sqrt{y(\beta-y)})$. For $V \in [0,1]$ and $\alpha,\gamma > 0$ let $\mathcal{B}_V(\alpha,\gamma) := B(\alpha,\gamma)^{-1}\int_0^V s^{\alpha-1}(1-s)^{\gamma-1}\,\mathrm{d}s$ be the regularised incomplete beta function.

Theorem 1.3. *Let $p > 0$ and $c = c(p)$. Let μ^* be the probability measure on $(0,\infty)$*

$$\mu^* := \int_0^1 \omega_{4ct^{-p}}\,\mathrm{d}t,$$

and let μ^*_u be its image under $x \mapsto x^{-1/p}$. Then:

1. *$\mu^*_u \in \mathcal{A}$, with density $\rho^*(u) = \mathcal{B}_{\min(1,\,u^{-p}/4c)}\bigl(\frac{1}{p}+\frac{1}{2},\frac{1}{2}\bigr)$. In particular $\rho^* = 1$ on $(0,u_s]$, where $u_s := (4c)^{-1/p} = 2I(p)/\pi$, and $0 < \rho^* < 1$ on (u_s,∞).*
2. *The logarithmic potential of μ^* satisfies $\int \log|x-y|\,\mathrm{d}\mu^*(y) = \log c + p$ for $0 < x \leq 4c$, and $\geq \log c + p$ for all $x > 0$.*
3. *$\mathcal{J}_p(\mu^*_u) = \kappa_p$, and $\mathcal{J}_p(\mu) < \kappa_p$ for every $\mu \in \mathcal{A}$ with $\mu \neq \mu^*_u$. Thus $\sup_{\mathcal{A}} \mathcal{J}_p = \kappa_p$, attained only at μ^*_u.*

Applied to the measure $\sum_k k^{-(2a+b)}\delta_{k^{-a}}$, Theorem 1.1 gives the asymptotics of Hankel determinants of zeta values along arithmetic progressions.

Corollary 1.4. *Let $a > 0$ and b be real numbers with $2a + b > 1$. Then*

$$\log \det_{1\leq i,j\leq n} \zeta\bigl(a(i+j)+b\bigr) = -an^2\log n + \Bigl(\frac{3a}{2} + \log c(a)\Bigr)n^2 + o(n^2).$$

In particular, $\log\det_{1\leq i,j\leq n}\zeta(i+j+b) = -n^2\log n + (\frac{3}{2} - \log 2)n^2 + o(n^2)$ for every $b > -1$, and $\log\det_{1\leq i,j\leq n}\zeta(2(i+j)+b) = -2n^2\log n + (3 + \log\frac{\pi^2}{16})n^2 + o(n^2)$ for every $b > -3$.

1.2. **Context.** *Hankel determinants of zeta values.* Monien [12] studied the Hankel determinants $H_n^{(r)}[F] = \det(F(i+j+r))$ of Dirichlet series F, gave a multiple-sum formula for them [12, Thm. 2.1], and wrote for $F = \zeta$: "We do not know how to prove the full asymptotic expansions but will describe a method that lets one at least understand the weaker asymptotic form $\log H_n^{(0)}[\zeta] \sim \log H_n^{(1)}[\zeta] \sim -n^2(\log(2n) - 3/2)$." He also reports numerical experiments of D. Zagier consistent with this form. Haynes and Zudilin [8] wrote: "for positive integers a and b, we expect that $\log\det_{1\le i,j\le n} \zeta(a(i+j)+b) = -an^2\log n + O(n^2)$ as $n\to\infty$", and proved the weaker bound "$\log\det_{1\le i,j\le n} \zeta(a(i+j)+b) \le -(a/2)n^2\log n + O(n^2)$". They also write that "In [5] Monien developed these ideas further and rigorously justified the above asymptotics in the case $r = 0$", their reference [5] being a personal communication [8, §1]. Coons and Sutherland [3, Thm. 1.3] proved, for every nondegenerate Dirichlet series $F(s) = \sum f(n)n^{-s}$ with $f \ge 0$ convergent for $\Re s \ge s_0$, that $H_n^{(r)}[F] > 0$ for $n \ge 1$, $r \ge s_0 - 2$, and that $\log H_n^{(r)}[F] < -cn^2$ for some $c > 0$ and all sufficiently large n and r.

Corollary 1.4 therefore proves the asymptotic formula expected in [8], for all positive integers a, b and more generally for real $a > 0$ and $2a + b > 1$, and it identifies the n^2 term. For $a = 1$ the constant $\frac{3}{2} - \log 2$ agrees with Monien's weak asymptotic form, since $-n^2(\log 2n - \frac{3}{2}) = -n^2\log n + (\frac{3}{2} - \log 2)n^2$. For $a = 2$ and $b \in \{-2, 0, 2\}$ the determinants are known in closed form, by results of Baricz and Štampach [1] on Hankel determinants of Rayleigh functions; their asymptotics are consistent with Corollary 1.4 (Remark 7.2). Apart from these closed-form cases, we have not found in the literature a published proof of the leading term $-an^2\log n$ (for $a = 1$, $r = 0$ see the remark of [8] quoted above), nor of the n^2 term $\frac{3a}{2} + \log c(a)$.

Hankel determinants of zeta-type moments also occur in recent irrationality work: Lai [11] writes that "Fauzan's proof uses Hankel determinants, which were previously studied by Haynes–Zudilin [23] and Zudilin [55]" (see [7]). The present paper is not concerned with arithmetic applications, and its results are logically independent of those works.

The equilibrium problem. The maximisation of $\mathcal{J}_p$ over $\mathcal{A}$ is a logarithmic energy problem with an upper constraint on the measure. Constrained energy problems of this kind were introduced by Rakhmanov [13] to describe the zero distribution of discrete orthogonal polynomials on a compact interval, and were extended by Dragnev and Saff [6] and Kuijlaars and Van Assche [10] (see the summary in [4]). Dragnev and Saff prove existence and uniqueness of the constrained extremal measure, together with a characterisation theorem and a saturation result, with explicit results for Krawtchouk polynomials; Kuijlaars and Van Assche show that extremal polynomials on infinite discrete sets, such as those of the Meixner and Charlier polynomials, are governed by a constrained extremal energy problem that can be solved explicitly. Van Assche and Van Baelen [14] treat a q-lattice $\{q^k\}$, which accumulates at 0. We use no result from this theory: existence, uniqueness and the value of the maximum are proved directly in Section 3. We have not found in the literature the problem $\sup_{\mathcal{A}} \mathcal{J}_p$, whose constraint has infinite total mass near the accumulation point, or its solution.

For $p = 1$, Monien [12, §3] set up a constrained maximisation of this type heuristically: "We seek a continuous test function $\rho(x)$ which maximizes the functional Eq. 3.5 subject to the constraint that $0 < \rho(x) \le 1$ for all x and $\rho(x) = 0$ for $x < 0$ and Eq. 3.4. We assume that such a continuous functions exists." His Theorem 3.1 gives the solution of the stationarity condition,

$$\rho(x) = 1 \quad (0 \le x \le \tfrac{1}{2}), \qquad \rho(x) = \tfrac{2}{\pi}\Big(\arctan\tfrac{1}{\sqrt{2x-1}} - \tfrac{\sqrt{2x-1}}{2x}\Big) \quad (x > \tfrac{1}{2}),$$

which coincides with the density ρ^* of Theorem 1.3 for $p = 1$ (Remark 3.6). His argument assumes existence, checks only stationarity and does not contain an upper bound or a discretisation step. Theorem 1.3 proves optimality and uniqueness for every $p > 0$, and Sections 4–5 supply the passage from the continuous problem to the determinants.

Averaged arcsine laws. The measure μ^* is an average of arcsine laws. A theorem of Kuijlaars and Van Assche on recurrence coefficients that vary regularly with n/N ([9], quoted from [5, Thm. 1.3]) produces measures of exactly this shape as limiting zero distributions. With their conventions, a family of Jacobi matrices whose diagonal and off-diagonal entries (after rescaling by N^p) behave like $2c\,t^{-p}$ and $c\,t^{-p}$ as $n/N \to t$ has, at $t = 1$, the limiting zero distribution $\int_0^1 \omega_{[0,4ct^{-p}]}\,\mathrm{d}t$. Heuristically, therefore, μ^* is the zero distribution that the measures of Theorem 1.1 would have if their recurrence coefficients satisfied $a_n \sim 2b_n \sim 2c(p)Cn^{-p}$ pointwise; the saturation condition in Theorem 1.3(i) is exactly the equation that fixes $c = c(p)$. This heuristic is not used in the proofs, and we do not prove pointwise asymptotics of the recurrence coefficients: Corollary 1.2 is a statement about double Cesàro means only.

1.3. **Method.** By Heine's formula (Lemma 2.1), D_n is a sum over n-subsets S of atoms of $\prod_{k\in S} w_k$ times a squared Vandermonde product. For $x_k = k^{-p}$ and $S = \{k_1 < \cdots < k_n\}$, the Vandermonde product is $n^{-pn(n-1)} \prod_{i\neq j} |g(u_i) - g(u_j)|$ with $g(u) = u^{-p}$ and $u_i = k_i/n$, and the empirical measure $\frac{1}{n}\sum \delta_{u_i}$ has at most one atom in each interval of length $1/n$, a discrete form of the constraint $\mu \le \mathrm{Leb}$. The upper bound (Section 4) compares point configurations with step densities (Lemma 4.1) and removes atoms beyond Mn by a tail estimate; the lower bound (Section 5) evaluates a single term of the Heine sum at the quantiles of a truncation of μ_u^*. Both bounds meet at $\sup_{\mathcal{A}} \mathcal{J}_p$, which Theorem 1.3 identifies. General regularly spaced atoms are reduced to $x_k = Ck^{-p}$ by a mean-value comparison (Lemma 2.3).

The only potential-theoretic input beyond elementary calculus is the positivity of the logarithmic energy of signed measures of total mass zero (Lemma 3.3), which is used only for the optimality and uniqueness part of Theorem 1.3(iii). It is a special case of a lemma of Beckermann, Kalyagin, Matos and Wielonsky [2, Lemma 2.1]; we include a short Fourier-analytic proof so that the paper is self-contained (Remark 3.4).

1.4. **Organisation and notation.** Section 2 collects the facts about Hankel determinants that we use. Section 3 proves Theorem 1.3. Sections 4 and 5 prove the upper and lower bounds for $x_k = k^{-p}$, and Section 6 completes the proofs of Theorem 1.1 and Corollary 1.2. Section 7 treats zeta values, and Section 8 reports numerical illustrations.

Throughout, $p > 0$ is fixed and $g(u) := u^{-p}$ on $(0, \infty)$. Constants denoted C_M, C_U, ... depend only on p and on the indicated parameters, and may change from line to line. We use the convention $0 \log 0 = 0$.

2. Hankel determinants of discrete measures

In this section $\sigma = \sum_{k\ge1} w_k \delta_{x_k}$ with $w_k \ge 0$, $\sum_k w_k < \infty$ and $\sup_k |x_k| < \infty$, so that all moments are finite. For a finite set $S \subset \mathbb{N}$ write

$$w(S) := \prod_{k\in S} w_k, \qquad V_x(S) := \prod_{\substack{k,l\in S\\ k<l}} (x_k - x_l)^2.$$

Lemma 2.1 (Heine's formula)**.** *For every $n \ge 1$,*

$$D_n(\sigma) = \sum_{\substack{S\subset\mathbb{N}\\ |S|=n}} w(S)\, V_x(S).$$

In particular all terms are nonnegative.

Proof. Write $m_{i+j} = \sum_k w_k x_k^i x_k^j$. Expanding the determinant multilinearly in its columns (legitimate because $\sum_k w_k |x_k|^r \le W X^r$ with $W = \sum_k w_k$ and $X := \sup_k |x_k|$, so all series converge absolutely),

$$D_n = \sum_{k_0,\dots,k_{n-1}\ge 1} \Big(\prod_{j=0}^{n-1} w_{k_j} x_{k_j}^j\Big) \det\big(x_{k_j}^i\big)_{i,j=0}^{n-1}.$$

The Vandermonde determinant vanishes unless the k_j are distinct. Group the remaining tuples according to the set $S = \{l_0 < \cdots < l_{n-1}\}$ they enumerate, $k_j = l_{\pi(j)}$ with π a permutation. Then $\det(x_{k_j}^i) = \operatorname{sgn}(\pi)\det(x_{l_j}^i)$ and

$$\sum_\pi \operatorname{sgn}(\pi) \prod_j x_{l_{\pi(j)}}^j = \det\big(x_{l_i}^j\big),$$

so the contribution of S is $w(S)\det(x_{l_j}^i)^2 = w(S)V_x(S)$. □

Lemma 2.2. *Let σ, σ' be measures as above.*

1. *(Monotonicity) If $\sigma' = \sum w'_k \delta_{x_k}$ with $w_k \le w'_k$ for all k, then $D_n(\sigma) \le D_n(\sigma')$.*
2. *(Translation) $D_n(\sum w_k \delta_{x_k+t}) = D_n(\sigma)$ for every $t \in \mathbb{R}$.*
3. *(Scaling) $D_n(\sum w_k \delta_{\lambda x_k}) = \lambda^{n(n-1)} D_n(\sigma)$ for $\lambda > 0$, and $D_n(\sum c w_k \delta_{x_k}) = c^n D_n(\sigma)$ for $c > 0$.*
4. *(Positivity) If $w_k > 0$ for infinitely many k with pairwise distinct x_k, then $D_n(\sigma) > 0$ for all n.*

Proof. All four statements follow from Lemma 2.1: its terms are nonnegative and increasing in each w_k; $V_x(S)$ depends only on differences; $V_{\lambda x}(S) = \lambda^{n(n-1)} V_x(S)$ and $(cw)(S) = c^n w(S)$; and under the hypothesis of (iv) some term is positive. □

Lemma 2.3 (Regular spacing comparison)**.** *Let $f, h \in C^1[1,\infty)$ with $f' < 0$ and $h' < 0$ on $[1,\infty)$, both bounded on $[1,\infty)$, and suppose that $\rho := f'/h'$ satisfies $R^{-1} \le \rho \le R$ on $[1,\infty)$ for some $R \ge 1$ and $\rho(\xi) \to 1$ as $\xi \to \infty$. Put $\eta(m) := \sup_{\xi \ge m} |\log \rho(\xi)|$. Then for all weights $w_k \ge 0$ with $\sum w_k < \infty$ and all $n \ge 1$,*

$$\Big|\log D_n\Big(\sum_k w_k \delta_{f(k)}\Big) - \log D_n\Big(\sum_k w_k \delta_{h(k)}\Big)\Big| \le 2\sum_{i=1}^{n-1} (n-i)\,\eta(i) = o(n^2),$$

whenever one (hence both) of the determinants is positive.

Proof. Let $S = \{k_1 < \cdots < k_n\}$. For $i < j$, Cauchy's mean value theorem gives $(f(k_i) - f(k_j))/(h(k_i) - h(k_j)) = \rho(\xi)$ for some $\xi \in (k_i, k_j)$. Since $\xi > k_i \ge i$, the ratio of the squared factors lies in $[e^{-2\eta(i)}, e^{2\eta(i)}]$. Multiplying over pairs, $V_f(S)/V_h(S)$ lies in $[e^{-E_n}, e^{E_n}]$ with $E_n = 2\sum_{i<n}(n-i)\eta(i)$, and the bound follows by comparing the Heine sums term by term. Finally η is bounded by $\log R$ and tends to 0, so $E_n = o(n^2)$. □

Lemma 2.4 (Jacobi coefficients)**.** *Suppose $D_n(\sigma) > 0$ for all n. Let P_k be the monic orthogonal polynomials of σ and $h_k := \int P_k^2 \,\mathrm{d}\sigma$. Then $D_n = \prod_{k=0}^{n-1} h_k$. Moreover $xP_k = P_{k+1} + a_{k+1}P_k + b_k^2 P_{k-1}$ for $k \ge 1$, with $b_k := (h_k/h_{k-1})^{1/2} > 0$, and*

$$D_n(\sigma) = m_0^n \prod_{j=1}^{n-1} b_j^{2(n-j)}.$$

The numbers b_k are the off-diagonal entries of the Jacobi matrix of σ (the coefficients of the three-term recurrence for the orthonormal polynomials $P_k/\sqrt{h_k}$).

Proof. The Hankel matrix $H_n = (m_{i+j})_{i,j<n}$ is the Gram matrix of $1, x, \dots, x^{n-1}$ in $L^2(\sigma)$. Writing $(P_0, \dots, P_{n-1})^T = L(1, \dots, x^{n-1})^T$ with L unit lower triangular gives $LH_nL^T = \operatorname{diag}(h_0, \dots, h_{n-1})$, so $D_n = \prod_{k<n} h_k$. The polynomial $xP_k - P_{k+1}$ has degree $\le k$, and it is orthogonal to P_l for $l \le k-2$; its P_{k-1}-coefficient is

$$\frac{1}{h_{k-1}} \int x P_k P_{k-1} \, \mathrm{d}\sigma = \frac{1}{h_{k-1}} \int P_k \, (xP_{k-1}) \, \mathrm{d}\sigma = \frac{h_k}{h_{k-1}}.$$

Hence $h_k = m_0 b_1^2 \cdots b_k^2$ and $D_n = \prod_{k=0}^{n-1} m_0 \prod_{j \le k} b_j^2 = m_0^n \prod_{j=1}^{n-1} b_j^{2(n-j)}$. □

Lemma 2.5. $2\sum_{j=1}^{n-1}(n-j)\log j = n^2 \log n - \frac{3}{2}n^2 + O(n \log n)$.

Proof. Let $\phi(x) = (n-x)\log x$ on $[1, n]$. Then $|\phi'(x)| \le \log n + n/x$, so $\left|\sum_{j=1}^{n-1} \phi(j) - \int_1^n \phi\right| \le \sum_{j=1}^{n-1}(\log n + n/j) = O(n \log n)$, and $\int_1^n (n-x)\log x \, \mathrm{d}x = \frac{n^2}{2}\log n - \frac{3}{4}n^2 + O(n)$. □

3. The constrained equilibrium problem

3.1. **Energies.** For finite positive Borel measures μ, ν on $(0, \infty)$ with $\int \log(1+x)\, \mathrm{d}\mu < \infty$ and $\int \log(1+x)\, \mathrm{d}\nu < \infty$ put

$$\mathcal{L}(\mu, \nu) := \iint \log|x-y| \, \mathrm{d}\mu(x) \, \mathrm{d}\nu(y), \qquad \mathcal{L}(\mu) := \mathcal{L}(\mu, \mu).$$

Since $\log|x-y| \le \log(1+x) + \log(1+y)$ for $x, y > 0$, the positive part of the integrand is integrable, so $\mathcal{L}(\mu, \nu) \in [-\infty, \infty)$ is well defined. Let

$$\sigma_p(\, \mathrm{d}x) := \tfrac{1}{p}\, x^{-1-1/p} \, \mathrm{d}x \quad \text{on } (0, \infty),$$

the image of Lebesgue measure on $(0, \infty)$ under $g(u) = u^{-p}$. For $\tau > 0$ let $\mathcal{A}_\tau$ be the set of positive Borel measures on $(0, \infty)$ of mass τ with $\mu \le$ Leb; thus $\mathcal{A} = \mathcal{A}_1$. For $\mu \in \mathcal{A}_\tau$ the image $\mu^x := g_*\mu$ satisfies $\mu^x \le \sigma_p$, hence $\int \log(1+x)\, \mathrm{d}\mu^x \le \tau \log 2 + \int_1^\infty \log(1+x)\, \mathrm{d}\sigma_p < \infty$, and we define

$$\mathcal{J}_p(\mu) := \mathcal{L}(\mu^x) = \iint \log|g(u) - g(v)| \, \mathrm{d}\mu(u) \, \mathrm{d}\mu(v) \in [-\infty, \infty), \qquad E_p := \sup_{\mu \in \mathcal{A}} \mathcal{J}_p(\mu).$$

Lemma 3.1 (Scaling)**.** *For a measure μ on $(0, \infty)$ and $\tau > 0$ let $\mu_\tau(B) := \tau\mu(B/\tau)$. If μ has mass m and density $\rho \le 1$, then μ_τ has mass τm and density $\rho(\cdot/\tau) \le 1$, and $\mathcal{J}_p(\mu_\tau) = \tau^2(\mathcal{J}_p(\mu) - pm^2 \log\tau)$. In particular $\mu \mapsto \mu_\tau$ is a bijection $\mathcal{A} \to \mathcal{A}_\tau$ and $\sup_{\mathcal{A}_\tau} \mathcal{J}_p = \tau^2(E_p - p\log\tau)$.*

Proof. $\mu_\tau(B) = \tau \int_{B/\tau} \rho(u)\, \mathrm{d}u = \int_B \rho(v/\tau)\, \mathrm{d}v$. Since $g(\tau u) = \tau^{-p} g(u)$,

$$\mathcal{J}_p(\mu_\tau) = \tau^2 \iint \log|g(\tau u) - g(\tau v)| \, \mathrm{d}\mu \, \mathrm{d}\mu = \tau^2\big(\mathcal{J}_p(\mu) - pm^2 \log\tau\big).$$

□

3.2. **Arcsine potentials.**

Lemma 3.2. *Let $\beta > 0$ and $U_\beta(x) := \int \log|x-y|\,\mathrm{d}\omega_\beta(y)$. Then $U_\beta(x) = \log(\beta/4)$ for $x \in [0,\beta]$, and U_β is increasing on $[\beta,\infty)$; in particular $U_\beta \geq \log(\beta/4)$ on $[0,\infty)$. Moreover, for every $x \geq 0$,*

$$\int \big|\log|x-y|\big|\,\mathrm{d}\omega_\beta(y) \leq 2\log(1+x+\beta) + \big|\log(\beta/4)\big|. \tag{3.1}$$

Proof. Write $x = \frac{\beta}{2}(1-\cos\varphi)$ and $y = \frac{\beta}{2}(1-\cos\theta)$ with $\theta,\varphi \in [0,\pi]$. Under $y = \frac{\beta}{2}(1-\cos\theta)$, the measure ω_β is the image of $\mathrm{d}\theta/\pi$ on $[0,\pi]$, since $\sqrt{y(\beta-y)} = \frac{\beta}{2}\sin\theta$. Now $\log|x-y| = \log\frac{\beta}{4} + \log|2(\cos\theta-\cos\varphi)|$, and

$$\log|2(\cos\theta-\cos\varphi)| = \log\Big|2\sin\tfrac{\theta+\varphi}{2}\Big| + \log\Big|2\sin\tfrac{\theta-\varphi}{2}\Big| = -\sum_{k\geq 1}\frac{2}{k}\cos k\theta\cos k\varphi,$$

using $\log|2\sin(\alpha/2)| = -\sum_{k\geq1} k^{-1}\cos k\alpha$, the Fourier series of an $L^2(0,2\pi)$ function. Each of the two series converges in $L^2([0,\pi],\mathrm{d}\theta)$, hence in L^1, so we may integrate termwise; since $\int_0^\pi \cos k\theta\,\mathrm{d}\theta = 0$ for $k \geq 1$, we get $U_\beta(x) = \log(\beta/4)$ for $x \in [0,\beta]$. For $x > \beta$, $U_\beta'(x) = \int (x-y)^{-1}\,\mathrm{d}\omega_\beta(y) > 0$, and U_β is continuous at β by dominated convergence (near $y=\beta$ the integrand is dominated by a multiple of $|\log s|\,s^{-1/2}$, $s = \beta - y$). For (3.1), write $|t| = 2t^+ - t$: since $|x-y| \leq x+\beta$ on the support, $\int|\log|x-y||\,\mathrm{d}\omega_\beta = 2\int\log^+|x-y|\,\mathrm{d}\omega_\beta - U_\beta(x) \leq 2\log(1+x+\beta) - \log(\beta/4)$. □

3.3. **Positivity of the energy of signed measures.**

Lemma 3.3. *Let ν be a finite signed Borel measure on $\mathbb{R}$ with $\nu(\mathbb{R}) = 0$ and $\iint|\log|x-y||\,\mathrm{d}|\nu|(x)\,\mathrm{d}|\nu|(y) < \infty$. Then*

$$-\iint \log|x-y|\,\mathrm{d}\nu(x)\,\mathrm{d}\nu(y) = \int_0^\infty \frac{|\hat\nu(\xi)|^2}{\xi}\,\mathrm{d}\xi \;\geq 0, \qquad \hat\nu(\xi) := \int e^{i\xi x}\,\mathrm{d}\nu(x),$$

with equality if and only if $\nu = 0$.

Proof. For $0 < \varepsilon < 1 < R$ and $a \in \mathbb{R}\setminus\{0\}$ put $K_{\varepsilon,R}(a) := \int_\varepsilon^R(\cos(a\xi) - e^{-\xi})\,\xi^{-1}\,\mathrm{d}\xi$.

Step 1: a uniform bound. We use three elementary estimates: (a) for $1 \leq A < B$, $|\int_A^B \cos t\,\frac{\mathrm{d}t}{t}| \leq 3/A \leq 3$ (integrate by parts); (b) for $0 < A < B \leq 1$, $\int_A^B \cos t\,\frac{\mathrm{d}t}{t} = \log\frac{B}{A} - \theta$ with $0 \leq \theta \leq \int_A^B \frac{t}{2}\,\mathrm{d}t \leq \frac{B^2}{4} \leq \frac14$, because $0 \leq 1-\cos t \leq t^2/2$; (c) $\int_\varepsilon^R e^{-\xi}\frac{\mathrm{d}\xi}{\xi} = \log\frac{1}{\varepsilon} + \theta'$ with $|\theta'| \leq 2$. Substitute $t = |a|\xi$, so that $\int_\varepsilon^R \cos(a\xi)\frac{\mathrm{d}\xi}{\xi} = \int_{\varepsilon|a|}^{R|a|}\cos t\,\frac{\mathrm{d}t}{t}$. If $\varepsilon|a| < 1 < R|a|$, then $K_{\varepsilon,R}(a) = -\log|a| + O(1)$. If $R|a| \leq 1$, then $K_{\varepsilon,R}(a) = \log R + O(1)$ and $0 < \log R \leq \log(1/|a|)$. If $\varepsilon|a| \geq 1$, then $K_{\varepsilon,R}(a) = -\log(1/\varepsilon) + O(1)$ and $0 < \log(1/\varepsilon) \leq \log|a|$. In all cases

$$|K_{\varepsilon,R}(a)| \leq \big|\log|a|\big| + 6 \qquad (0<\varepsilon<1<R,\ a\neq 0). \tag{3.2}$$

Step 2: the pointwise limit. We have $K_{\varepsilon,R}(a) - K_{\varepsilon,R}(1) = \int_{\varepsilon|a|}^{\varepsilon}\cos t\,\frac{\mathrm{d}t}{t} + \int_R^{R|a|}\cos t\,\frac{\mathrm{d}t}{t}$. As $\varepsilon \to 0$ the first integral tends to $\log(1/|a|)$, by (b); as $R\to\infty$ the second tends to 0, by (a). Also $K_{\varepsilon,R}(1) \to k_0 := \int_0^\infty(\cos\xi - e^{-\xi})\xi^{-1}\,\mathrm{d}\xi$, which converges. Hence $K_{\varepsilon,R}(a) \to k_0 - \log|a|$.

Step 3. The integrability hypothesis forces $|\nu|\otimes|\nu|$ to give zero mass to the diagonal $\{x=y\}$. By (3.2) and dominated convergence (with respect to $|\nu|\otimes|\nu|$, dominating function $|\log|x-y|| + 6$), and since $\nu(\mathbb{R}) = 0$,

$$\iint K_{\varepsilon,R}(x-y)\,\mathrm{d}\nu\,\mathrm{d}\nu \longrightarrow k_0\,\nu(\mathbb{R})^2 - \iint\log|x-y|\,\mathrm{d}\nu\,\mathrm{d}\nu = -\iint\log|x-y|\,\mathrm{d}\nu\,\mathrm{d}\nu.$$

On the other hand, the integrand $(\xi,x,y)\mapsto(\cos((x-y)\xi)-e^{-\xi})/\xi$ is bounded on $[\varepsilon,R]\times\mathbb{R}^2$, so by Fubini, $\iint\cos((x-y)\xi)\,\mathrm{d}\nu\,\mathrm{d}\nu=|\hat\nu(\xi)|^2$ and $\nu(\mathbb{R})=0$,

$$\iint K_{\varepsilon,R}(x-y)\,\mathrm{d}\nu\,\mathrm{d}\nu=\int_\varepsilon^R\frac{|\hat\nu(\xi)|^2}{\xi}\,\mathrm{d}\xi.$$

Letting $\varepsilon\to0$, $R\to\infty$ (monotone convergence on the right) gives the identity.

Equality. If the integral vanishes, then $\hat\nu=0$ on $(0,\infty)$ by continuity, hence on $\mathbb{R}$, because $\hat\nu(-\xi)=\overline{\hat\nu(\xi)}$ and $\hat\nu(0)=\nu(\mathbb{R})=0$. Then $\nu=0$: for the Gaussian $G_t(x)=(2\pi t)^{-1/2}e^{-x^2/2t}$ we have $G_t(x)=\frac{1}{2\pi}\int e^{-t\xi^2/2}e^{-i\xi x}\,\mathrm{d}\xi$, so by Fubini $(G_t*\nu)(x)=\frac{1}{2\pi}\int e^{-t\xi^2/2}e^{-i\xi x}\hat\nu(\xi)\,\mathrm{d}\xi=0$ for all x and $t>0$; for bounded continuous ψ, $\int\psi\,\mathrm{d}\nu=\lim_{t\to0}\int(G_t*\psi)\,\mathrm{d}\nu=\lim_{t\to0}\int\psi\,(G_t*\nu)\,\mathrm{d}x=0$ by dominated convergence, so $\nu=0$. □

Remark 3.4. Lemma 3.3 is a special case of [2, Lemma 2.1]. There the energy is $I(\mu)=\iint\log\frac{1}{|x-y|}\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y)$, $M(\Sigma)$ denotes the Borel measures of finite mass supported in Σ with finite energy and with $\int\log(1+|t|)\,\mathrm{d}\mu(t)<\infty$ [2, (1.12)], and the statement reads: "Lemma 2.1. Let $\mu,\nu\in M(\mathbb{C})$ with $\|\mu\|=\|\nu\|$. Then $I(\mu-\nu)\ge0$, and $I(\mu-\nu)=0$ if and only if $\mu=\nu$." The proof given there refers to the literature for both the compactly supported and the unbounded case. In the application below ($\mu=g_*\mu_0$ with $\mu_0\in\mathcal{A}$ and $\nu=\mu^*$), both measures are positive, of mass 1, of finite energy, and satisfy $\int\log(1+x)<\infty$, so they lie in $M(\mathbb{C})$ and [2, Lemma 2.1] applies as well. Either reference suffices; we use Lemma 3.3.

3.4. **The explicit maximiser.** Throughout this subsection $c=c(p)$ and $\beta_t:=4ct^{-p}$ for $t\in(0,1]$, so that $\mu^*=\int_0^1\omega_{\beta_t}\,\mathrm{d}t$. Put $U^*(x):=\int\log|x-y|\,\mathrm{d}\mu^*(y)$.

Proof of Theorem 1.3(i). Fix $x>0$ and let $t_x:=(4c/x)^{1/p}$, so that $x<\beta_t$ if and only if $t<t_x$. Then

$$\frac{\mathrm{d}\mu^*}{\mathrm{d}x}=\frac{1}{\pi}\int_0^{\min(1,t_x)}\frac{\mathrm{d}t}{\sqrt{x(\beta_t-x)}}=\frac{t_x}{\pi x}\int_0^{\min(1,1/t_x)}\frac{\mathrm{d}s}{\sqrt{s^{-p}-1}},$$

by the substitution $t=t_xs$, for which $\beta_t=xs^{-p}$. The substitution $v=s^p$ gives, for $0\le V\le1$,

$$\int_0^{V^{1/p}}\frac{\mathrm{d}s}{\sqrt{s^{-p}-1}}=\frac{1}{p}\int_0^V v^{\frac1p-\frac12}(1-v)^{-\frac12}\,\mathrm{d}v=\frac{1}{p}B\big(\tfrac1p+\tfrac12,\tfrac12\big)\mathcal{B}_V\big(\tfrac1p+\tfrac12,\tfrac12\big),$$

and $B(\frac1p+\frac12,\frac12)=\sqrt{\pi}\,\Gamma(\frac1p+\frac12)/\Gamma(\frac1p+1)=2I(p)$. With $V=\min(1,t_x^{-p})=\min(1,x/4c)$ we obtain

$$\frac{\mathrm{d}\mu^*}{\mathrm{d}x}=\frac{(4c)^{1/p}x^{-1-1/p}}{\pi}\cdot\frac{2I(p)}{p}\cdot\mathcal{B}_{\min(1,x/4c)}\big(\tfrac1p+\tfrac12,\tfrac12\big)=\mathcal{B}_{\min(1,x/4c)}\big(\tfrac1p+\tfrac12,\tfrac12\big)\cdot\frac{\mathrm{d}\sigma_p}{\mathrm{d}x},$$

because $(4c)^{1/p}=\pi/(2I(p))$ by the definition of $c=c(p)$. Thus $\mu^*=\sigma_p$ on $[4c,\infty)$ and $\mu^*<\sigma_p$ (in density) on $(0,4c)$. Pulling back by $x=u^{-p}$, which maps σ_p to Lebesgue measure, gives the density ρ^* of μ_u^*; ρ^* is continuous and positive on $(0,\infty)$, equal to 1 exactly on $(0,u_s]$, and $u_s=(4c)^{-1/p}=2I(p)/\pi$. The total mass is $\int_0^1\mathrm{d}t=1$. Hence $\mu_u^*\in\mathcal{A}$. □

Proof of Theorem 1.3(ii). By (3.1), for fixed $x>0$, $\int_0^1\int|\log|x-y||\,\mathrm{d}\omega_{\beta_t}(y)\,\mathrm{d}t\le\int_0^1\big(2\log(1+x+\beta_t)+|\log(\beta_t/4)|\big)\,\mathrm{d}t<\infty$, since $\log\beta_t=\log4c-p\log t$. By Fubini, $U^*(x)=\int_0^1U_{\beta_t}(x)\,\mathrm{d}t$. If $x\le4c=\beta_1$, then $x\le\beta_t$ for all $t\in(0,1]$, and Lemma 3.2 gives $U^*(x)=\int_0^1\log(ct^{-p})\,\mathrm{d}t=\log c+p$. For all $x>0$, Lemma 3.2 gives $U_{\beta_t}(x)\ge\log(\beta_t/4)$, hence $U^*(x)\ge\log c+p$. □

Proof of Theorem 1.3(iii). The value. For $0 < s \le t \le 1$ we have $\operatorname{supp}\omega_{\beta_t} \subset [0,\beta_t] \subset [0,\beta_s]$, so by Lemma 3.2

$$\iint \log|x-y|\,\mathrm{d}\omega_{\beta_s}(x)\,\mathrm{d}\omega_{\beta_t}(y) = \int U_{\beta_s}\,\mathrm{d}\omega_{\beta_t} = \log\frac{\beta_s}{4} = \log c - p\log\min(s,t).$$

By (3.1), $\iint |\log|x-y||\,\mathrm{d}\omega_{\beta_s}\,\mathrm{d}\omega_{\beta_t} \le 2\log(1+\beta_s+\beta_t) + |\log(\beta_t/4)|$, which is integrable over $(s,t)\in(0,1]^2$. Hence $\mathcal{L}(\mu^*)$ converges absolutely, and by Fubini

$$\begin{aligned}\mathcal{J}_p(\mu_u^*) = \mathcal{L}(\mu^*) &= \int_0^1\int_0^1 \big(\log c - p\log\min(s,t)\big)\,\mathrm{d}s\,\mathrm{d}t \\ &= \log c - p\int_0^1 2(1-m)\log m\,\mathrm{d}m = \log c + \frac{3p}{2} = \kappa_p.\end{aligned}$$

Optimality. Let $\mu\in\mathcal{A}$. If $\mathcal{J}_p(\mu) = -\infty$ there is nothing to prove, so assume $\mathcal{J}_p(\mu) > -\infty$ and let $\mu^x = g_*\mu \le \sigma_p$. We first check that all energies below converge absolutely.

- $\mathcal{L}(\mu^x)$: the positive part of the integrand is integrable (Section 3.1), and the negative part is integrable because $\mathcal{L}(\mu^x) > -\infty$.
- $\mathcal{L}(\mu^*)$: shown above.
- $\mathcal{L}(\mu^x,\mu^*)$: using $|t| = 2t^+ - t$, $\log^+|x-y| \le \log(1+x)+\log(1+y)$ and part (ii),

$$\iint \big|\log|x-y|\big|\,\mathrm{d}\mu^*(y)\,\mathrm{d}\mu^x(x) \le \int\Big(2\log(1+x)+2\int\log(1+y)\,\mathrm{d}\mu^*(y) - (\log c+p)\Big)\,\mathrm{d}\mu^x(x) < \infty,$$

since $\mu^* \le \sigma_p$ and $\mu^x \le \sigma_p$ have finite logarithmic moments and mass 1.

Let $\nu := \mu^x - \mu^*$, a finite signed measure on $(0,\infty)$ with $\nu(\mathbb{R}) = 0$ and $\iint|\log|x-y||\,\mathrm{d}|\nu|\,\mathrm{d}|\nu| < \infty$. Expanding bilinearly (all terms are finite),

$$\mathcal{J}_p(\mu) = \mathcal{L}(\mu^x) = \mathcal{L}(\mu^*) + 2\int U^*\,\mathrm{d}\nu + \iint\log|x-y|\,\mathrm{d}\nu\,\mathrm{d}\nu.$$

By Lemma 3.3 the last term is ≤ 0, with equality only if $\nu = 0$. For the middle term, ν has mass 0, so $\int U^*\,\mathrm{d}\nu = \int(U^* - \log c - p)\,\mathrm{d}\nu$. By (ii) the integrand vanishes on $(0,4c]$ and is ≥ 0 on $(4c,\infty)$, where $\nu = \mu^x - \sigma_p \le 0$ by (i). Hence $\int U^*\,\mathrm{d}\nu \le 0$, and $\mathcal{J}_p(\mu) \le \mathcal{L}(\mu^*) = \kappa_p$, with equality only if $\nu = 0$, that is, $\mu = \mu_u^*$. □

Remark 3.5. The only step of the paper that uses Lemma 3.3 (equivalently [2, Lemma 2.1]) is the optimality and uniqueness part of Theorem 1.3(iii). In the proof of Theorem 1.1 it is needed for the inequality $E_p \le \kappa_p$, which enters the upper bound. The lower bound uses only parts (i) and the value $\mathcal{J}_p(\mu_u^*) = \kappa_p$.

Remark 3.6. For $p = 1$ we have $c = \frac12$ and $\rho^*(u) = \mathcal{B}_{\min(1,1/2u)}(\frac32,\frac12)$. We checked numerically (mpmath, 30 digits; script `checks/monien_check.py`) that this agrees with Monien's density ρ quoted in Section 1.2 at eleven points $x\in[0.1,10^4]$, to within 10^{-30}.

4. Discretisation and the upper bound

In Sections 4 and 5 we write, for a finite $S = \{k_1 < \cdots < k_N\} \subset \mathbb{N}$,

$$V(S) := \prod_{i<j}\big(k_i^{-p} - k_j^{-p}\big)^2, \qquad u_i := k_i/n.$$

Since $k^{-p} = n^{-p}g(k/n)$,

$$\log V(S) = -pN(N-1)\log n + \sum_{i\ne j}\log|g(u_i) - g(u_j)|. \tag{4.1}$$

Lemma 4.1 (Discretisation)**.** *Let $M \geq 1$. There is C_M such that for all $n \geq 2$ and all $S = \{k_1 < \cdots < k_N\} \subset \mathbb{N} \cap [1, Mn]$ with $N \leq n$,*

$$\sum_{i \neq j} \log |g(u_i) - g(u_j)| \leq n^2 \mathcal{J}_p(\bar{\mu}_S) + C_M n \log n,$$

where $\bar{\mu}_S \in \mathcal{A}_{N/n}$ has density 1 on $\bigcup_{k \in S} A_k$, $A_k := [(k-1)/n, k/n]$.

Proof. For cells A, A' write $\langle F \rangle_{A \times A'} := n^2 \iint_{A \times A'} F(u,v)\,\mathrm{d}u\,\mathrm{d}v$ for the average of $F(u,v) := \log|g(u) - g(v)|$. Then $n^2 \mathcal{J}_p(\bar{\mu}_S) = \sum_{i \neq j} \langle F \rangle_{A_{k_i} \times A_{k_j}} + \sum_i \langle F \rangle_{A_{k_i}^2}$. Recall that $|g'(u)| = pu^{-p-1}$ is decreasing.

Non-adjacent pairs. Let $k < l$ with $l \geq k+2$. For $u \in A_k$, $v \in A_l$ we have $g(u) - g(v) \geq g(k/n) - g((l-1)/n) =: D > 0$. With $\Delta := g((l-1)/n) - g(l/n) \leq |g'((l-1)/n)|/n$ and $D \geq \frac{l-1-k}{n} |g'((l-1)/n)|$ (mean value theorem and monotonicity of $|g'|$),

$$\log\big(g(k/n) - g(l/n)\big) = \log(D + \Delta) \leq \log D + \frac{\Delta}{D} \leq \langle F \rangle_{A_k \times A_l} + \frac{1}{l-k-1}.$$

Adjacent pairs. Let $l = k+1$. The point value is $\log(g(k/n) - g((k+1)/n)) \leq \log(|g'(k/n)|/n)$. For $u \in A_k$, $v \in A_{k+1}$, $g(u) - g(v) \geq |g'((k+1)/n)|(v-u)$, so $\langle F \rangle_{A_k \times A_{k+1}} \geq \log|g'((k+1)/n)| - \log n + c_0$ with $c_0 := \int_0^1 \int_1^2 \log(y-x)\,\mathrm{d}y\,\mathrm{d}x = 2\log 2 - \frac{3}{2}$. Since $|g'(k/n)|/|g'((k+1)/n)| = (1 + 1/k)^{p+1} \leq 2^{p+1}$, the point value exceeds the average by at most $(p+1)\log 2 - c_0$.

Diagonal cells. For $u, v \in A_k$, $|g(u) - g(v)| \geq |g'(k/n)||u - v|$, so $\langle F \rangle_{A_k^2} \geq \log|g'(k/n)| - \log n - \frac{3}{2}$, using $\iint_{[0,1]^2} \log|x - y| = -\frac{3}{2}$. For $k \leq Mn$, $|g'(k/n)| \geq pM^{-p-1}$.

Summation. For fixed i, the partners $j > i$ with $k_j \geq k_i + 2$ have distinct values $k_j - k_i - 1 \geq 1$, so $\sum_j 1/(k_j - k_i - 1) \leq 1 + \log n$; there is at most one adjacent partner. Counting ordered pairs,

$$\sum_{i \neq j} \log|g(u_i) - g(u_j)| \leq n^2 \mathcal{J}_p(\bar{\mu}_S) + 2N\big(1 + \log n + (p+1)\log 2 + |c_0|\big) + N\big(\log n + \tfrac{3}{2} + (p+1)\log M + |\log p|\big),$$

which is of the stated form since $N \leq n$. □

The diagonal estimate is the only place where $S \subset [1, Mn]$ is used: an isolated point $k \gg n$ makes $\mathcal{J}_p(\bar{\mu}_S)$ very negative while contributing little to the point sum. Atoms beyond Mn are therefore treated separately in Theorem 4.3.

Corollary 4.2. *For $M \geq 1$ there is C'_M such that for $n \geq 2$ and $S \subset \mathbb{N} \cap [1, Mn]$ with $|S| = N \leq n$,*

$$\log V(S) \leq -pN^2 \log N + E_p N^2 + C'_M n \log n.$$

Proof. For $N = 0$ both sides are trivial. Otherwise, by (4.1), Lemma 4.1 and Lemma 3.1 with $\tau = N/n$,

$$\log V(S) \leq -pN(N-1)\log n + N^2\big(E_p - p\log(N/n)\big) + C_M n \log n,$$

and $-pN(N-1)\log n - pN^2 \log(N/n) = -pN^2 \log N + pN \log n$. □

Note that only the value of the supremum E_p enters here; no information about maximisers is used.

Theorem 4.3 (Upper bound)**.** *Let $w_k \geq 0$ with $W = \sum_k w_k < \infty$ and $\sigma_0 := \sum_k w_k \delta_{k^{-p}}$. Then*

$$\limsup_{n\to\infty} \frac{1}{n^2}\Big(\log D_n(\sigma_0) + pn^2 \log n\Big) \leq E_p.$$

Proof. Fix $M \geq 2$. Each n-set $S \subset \mathbb{N}$ splits uniquely as $S = H \cup T$ with $H := S \cap [1, Mn]$ and $T := S \cap (Mn, \infty)$; let $m := |T|$ and $N := n - m = |H|$. Then $V(S) = V(H)V(T)\prod_{a\in H, b\in T}(a^{-p} - b^{-p})^2$.

Cross pairs. For $a \in H$ and $b \in T$, $0 < a^{-p} - b^{-p} < a^{-p}$. Since H consists of N distinct positive integers, $\prod_{a\in H} a \geq N!$, so the cross product is at most $(N!)^{-2pm}$.

Tail. All atoms b^{-p} with $b \in T$ lie in the interval $(0, (Mn)^{-p})$; hence

$$V(T) \leq (Mn)^{-pm(m-1)}, \qquad \sum_{|T|=m} w(T) \leq W^m.$$

Head. $\sum_{|H|=N} w(H)V(H) \leq W^N \sup_H V(H)$, and Corollary 4.2 bounds $\sup_H V(H)$.

By Lemma 2.1, $D_n(\sigma_0) \leq \sum_{m=0}^n T_m$, where

$$\log T_m \leq n \log^+ W - 2pm \log N! - pN^2 \log N + E_p N^2 - pm(m-1)\log(Mn) + C'_M n \log n.$$

Recall that $E_p = \kappa_p$ is finite (Theorem 1.3(iii)). Put $t := m/n$. If $N \geq 1$, use $\log N! \geq N \log N - N$, $\log N = \log n + \log(1-t)$ and $\log(Mn) = \log n + \log M$. The coefficient of $\log n$ is $-p(2mN + N^2 + m^2) = -pn^2$, and the remaining terms give

$$\log T_m + pn^2 \log n - E_p n^2 \leq n^2 \phi_M(t) + O_M(n \log n),$$

$$\phi_M(t) := -p(1-t)(1+t)\log(1-t) + 2pt(1-t) - E_p\, t(2-t) - pt^2 \log M.$$

Since $-(1-t)\log(1-t) \leq t$ on $[0,1)$, we get $\phi_M(t) \leq p(1+t)t + 2pt + 2|E_p|t - pt^2 \log M \leq At - pt^2 \log M$ with $A := 4p + 2|E_p|$. If $N = 0$ ($t = 1$), then directly $\log T_n \leq n \log^+ W - pn(n-1)\log(Mn) + C'_M n \log n$, and $-p \log M - E_p \leq A - p \log M$, so the same bound holds. The $O_M(n \log n)$ terms are uniform in m, and $\max_{t\in[0,1]}(At - pt^2 \log M) \leq A^2/(4p \log M)$. Summing the $n+1$ terms,

$$\limsup_{n\to\infty} \frac{1}{n^2}\Big(\log D_n(\sigma_0) + pn^2 \log n\Big) \leq E_p + \frac{A^2}{4p \log M}.$$

Let $M \to \infty$. □

5. The lower bound

Lemma 5.1. *Let φ be decreasing on $(0,\infty)$ and integrable near 0, and let $\lambda \leq \mathrm{Leb}$ be a measure on $(0,\infty)$ of mass $\delta > 0$ with $\int |\varphi|\,\mathrm{d}\lambda < \infty$. Then $\int \varphi\,\mathrm{d}\lambda \leq \int_0^\delta \varphi(u)\,\mathrm{d}u$.*

Proof. $\int \varphi\,\mathrm{d}\lambda - \int_0^\delta \varphi = \int_{(0,\delta]} \varphi\,\mathrm{d}(\lambda - \mathrm{Leb}) + \int_{(\delta,\infty)} \varphi\,\mathrm{d}\lambda \leq \varphi(\delta)\big(\lambda((0,\delta]) - \delta\big) + \varphi(\delta)\lambda((\delta,\infty)) = 0$, because $\lambda - \mathrm{Leb} \leq 0$ and $\varphi \geq \varphi(\delta)$ on $(0,\delta]$, and $\varphi \leq \varphi(\delta)$ on (δ,∞). □

Theorem 5.2 (Lower bound)**.** *Let $w_k \geq c_1 k^{-s}$ for all k, with $c_1 > 0$, $s \in \mathbb{R}$, and let $\sigma_0 = \sum_k w_k \delta_{k^{-p}}$ (with $\sum w_k < \infty$, so that the moments are finite). Then*

$$\liminf_{n\to\infty} \frac{1}{n^2}\Big(\log D_n(\sigma_0) + pn^2 \log n\Big) \geq \kappa_p.$$

Proof. Fix $\eta > 0$.

Step 1: a compactly supported admissible approximant. By Theorem 1.3(i), μ_u^* has a continuous positive density $\rho^* \leq 1$ on $(0,\infty)$. For $U > 0$ let $\mu_U := \mu_u^*|_{(0,U]}$, of mass m_U. By the proof of Theorem 1.3(iii), $|\log|g(u) - g(v)||$ is $\mu_u^* \otimes \mu_u^*$-integrable, so by dominated convergence

$\mathcal{J}_p(\mu_U) \to \mathcal{J}_p(\mu_u^*) = \kappa_p$ and $m_U \to 1$ as $U \to \infty$. Let $\theta := (\mu_U)_{1/m_U}$ (notation of Lemma 3.1). Then θ has mass 1, density $\rho^*(um_U) \in [d_U, 1]$ on its support $(0, U']$, where $U' := U/m_U$ and $d_U := \min_{(0,U]} \rho^* \in (0,1]$, and density 0 beyond. So $\theta \in \mathcal{A}$, and by Lemma 3.1

$$\mathcal{J}_p(\theta) = m_U^{-2}\mathcal{J}_p(\mu_U) + p\log m_U \geq \kappa_p - \eta$$

for U large enough, which we fix. Constants C_U below depend on p and U only.

Step 2: the quantile configuration. Let F be the distribution function of θ; it is continuous and strictly increasing on $[0, U']$. For $1 \leq i \leq n$ let $B_i = [l_i, r_i] := [F^{-1}(\frac{i-1}{n}), F^{-1}(\frac{i}{n})]$, $q_i := F^{-1}(\frac{i-1/2}{n})$, $k_i := \lceil nq_i \rceil$ and $u_i := k_i/n \in [q_i, q_i + \frac{1}{n})$.

- Since $\theta \leq \text{Leb}$ and $\theta(B_i) = \frac{1}{n}$: $|B_i| \geq \frac{1}{n}$, so $l_i \geq \frac{i-1}{n}$ and $r_i \geq \frac{i}{n}$; also $r_i - q_i \geq \frac{1}{2n}$ and $q_i - l_i \geq \frac{1}{2n}$. Hence $q_{i+1} - q_i \geq \frac{1}{n}$, so the k_i are distinct, $k_1 \geq 1$, and $u_j - u_i \geq \frac{1}{n}$ for $i < j$. Moreover $u_i < q_i + \frac{1}{n} \leq r_i + \frac{1}{2n} \leq r_{i+1}$.
- Since the density is $\geq d_U$ on $(0, U']$: $|B_i| \leq \frac{1}{d_U n}$, so $r_i \leq \frac{i}{d_U n}$, and $u_j \leq r_j + \frac{1}{2n} \leq \frac{j+1}{d_U n}$.
- $S^* := \{k_1 < \cdots < k_n\} \subset [1, U'n + 1]$.

Step 3: energy comparison. Write $F(u,v) = \log|g(u) - g(v)|$ and, for blocks B, B', put $\langle F\rangle_{B\times B'} := n^2 \iint_{B\times B'} F \,\mathrm{d}\theta \,\mathrm{d}\theta$; this is an average, since $\theta(B_i) = \frac{1}{n}$. Then

$$n^2\mathcal{J}_p(\theta) = \sum_{i\neq j}\langle F\rangle_{B_i\times B_j} + \sum_i \langle F\rangle_{B_i^2}.$$

For $i < j$, $u \in B_i$ and $v \in B_j$ we have $u \leq v$ and

$$F(u,v) = -p\log u + h(u/v), \qquad h(r) := \log(1 - r^p),$$

with h decreasing on $(0,1)$. We show that $F(u_i, u_j) \geq \langle F\rangle_{B_i\times B_j} - \mathrm{err}_{ij}$ with $\sum_{i<j}\mathrm{err}_{ij} = O_U(n\log n)$. We use repeatedly that for $0 < r < 1$,

$$\min(1,p)(1-r) \leq 1 - r^p \leq \max(1,p)(1-r). \tag{5.1}$$

(a) *The term $-p\log u$.* The point value minus the average is $-p\langle\log(u_i/u)\rangle_{B_i}$, and $u_i \leq r_{i+1}$, $u \geq l_i$ on B_i. For $i \geq 2$, $\log(r_{i+1}/l_i) \leq (r_{i+1} - l_i)/l_i \leq \frac{2/(d_U n)}{(i-1)/n}$, so the loss is at most $2p/(d_U(i-1))$. For $i = 1$, Lemma 5.1 (with $\varphi = \log(1/u)$ and $\lambda = \theta|_{B_1}$, $\delta = 1/n$) gives $\langle\log(1/u)\rangle_{B_1} \leq n\int_0^{1/n}\log\frac{1}{u}\,\mathrm{d}u = \log n + 1$, and $\log u_1 \leq \log r_2 \leq \log\frac{2}{d_U n}$; so the loss is at most $p(\log\frac{2}{d_U} + 1)$. Summing over $j > i$ and then over i gives $O_U(n\log n)$.

(b) *The term h, for $j \geq i+3$.* On $B_i \times B_j$, $u/v \geq a := l_i/r_j$, so the average of $h(u/v)$ is at most $h(a)$. At the point, $u_i/u_j \leq b := r_{i+1}/l_j$, and $b < 1$ because $l_j = r_{j-1} \geq r_{i+2} > r_{i+1}$; so $h(u_i/u_j) \geq h(b)$. The loss is at most $h(a) - h(b) = \log\frac{1-a^p}{1-b^p} \leq \frac{b^p - a^p}{1-b^p}$. By (5.1), $b^p - a^p = b^p(1 - (a/b)^p) \leq \max(1,p)(1 - a/b)$.

- If $b \leq \frac{1}{2}$: $1 - b^p \geq 1 - 2^{-p}$, and $1 - \frac{a}{b} \leq \big(1 - \frac{l_i}{r_{i+1}}\big) + \big(1 - \frac{l_j}{r_j}\big) \leq \frac{2/(d_U n)}{(i+1)/n} + \frac{1/(d_U n)}{j/n} \leq \frac{3}{d_U(i+1)}$. The loss is at most $C_U/(i+1)$.
- If $b > \frac{1}{2}$: by (5.1), $1 - b^p \geq \min(1,p)(1-b) = \min(1,p)\frac{l_j - r_{i+1}}{l_j} \geq \min(1,p)\frac{j-i-2}{nl_j}$, since the $j-i-2$ blocks $B_{i+2}, \ldots, B_{j-1}$ each have length $\geq \frac{1}{n}$. Also $1 - \frac{a}{b} = \frac{b-a}{b} \leq 2(b-a)$ and $b - a \leq \frac{r_{i+1} - l_i}{l_j} + \frac{l_i}{r_j}\cdot\frac{r_j - l_j}{l_j} \leq \frac{3}{d_U n l_j}$. The loss is at most $C_U/(j-i-2)$.

Summing, $\sum_{j\geq i+3}\big(C_U/(i+1) + C_U/(j-i-2)\big) = O_U(n\log n)$.

(c) *The term h, for $j \in \{i+1, i+2\}$.* The average of h is ≤ 0 since $h \leq 0$. At the point, by (5.1), $1 - (u_i/u_j)^p \geq \min(1,p)\frac{u_j - u_i}{u_j} \geq \min(1,p)\frac{1/n}{(j+1)/(d_U n)} = \frac{\min(1,p)d_U}{j+1}$. So the loss is at most $\log(j+1) + C_U$, and there are at most $2n$ such pairs: $O_U(n\log n)$.

(d) *Diagonal blocks.* $|g(u)-g(v)| \le g(\min(u,v))$, so $\langle F\rangle_{B_i^2} \le p\langle \log(1/\min(u,v))\rangle_{B_i^2}$. For $i \ge 2$, $\min(u,v) \ge l_i \ge \frac{i-1}{n}$, giving $\le p\log\frac{n}{i-1}$. For $i=1$ and $n \ge 1/d_U$ we have $B_1 \subset (0,1]$, so $\log\frac{1}{\min(u,v)} \le \log\frac{1}{u} + \log\frac{1}{v}$, and Lemma 5.1 gives $\le 2p(\log n+1)$. In total $\sum_i \langle F\rangle_{B_i^2} \le p\big((n-1)\log n - \log(n-1)!\big) + 2p(\log n+1) = O(n)$.

Combining (a)–(d),

$$\sum_{i\ne j}\log|g(u_i)-g(u_j)| = 2\sum_{i<j}F(u_i,u_j) \ge n^2\mathcal{J}_p(\theta) - \sum_i \langle F\rangle_{B_i^2} - O_U(n\log n)$$
$$\ge n^2\mathcal{J}_p(\theta) - O_U(n\log n).$$

Step 4: conclusion. By Lemma 2.1, $D_n(\sigma_0) \ge w(S^*)V(S^*)$. Here $\log w(S^*) \ge n\log c_1 - |s|\,n\log(U'n+1)$, and by (4.1) and Step 3,

$$\log D_n(\sigma_0) \ge -pn(n-1)\log n + n^2(\kappa_p-\eta) - O_U(n\log n).$$

Let $n\to\infty$ and then $\eta\to 0$. □

The lower bound uses Theorem 1.3(i) and the value $\mathcal{J}_p(\mu_u^*) = \kappa_p$, but neither Lemma 3.3 nor any optimality statement.

6. Proof of Theorem 1.1 and Corollary 1.2

Proof of Theorem 1.1. Pure powers. Let $\sigma_0 = \sum_k w_k\delta_{k^{-p}}$ with weights satisfying (1.4). By Theorems 4.3, 5.2 and 1.3(iii) ($E_p = \kappa_p$), $\log D_n(\sigma_0) = -pn^2\log n + \kappa_p n^2 + o(n^2)$. By Lemma 2.2(iii), the measure $\sum_k w_k\delta_{Ck^{-p}}$ has $\log D_n$ larger by $n(n-1)\log C = n^2\log C + O(n)$.

General atoms. Let f satisfy (1.3) and put $h(\xi) := C\xi^{-p}$. Since f' is continuous, $\xi^{p+1}|f'(\xi)|$ is bounded below by a positive constant, and $\xi^{p+1}f'(\xi) \to -pC < 0$, we have $f' < 0$ on $[1,\infty)$; f is bounded because $|f'(\xi)| \le C'\xi^{-p-1}$ is integrable. The ratio $\rho = f'/h' = \xi^{p+1}f'(\xi)/(-pC)$ is bounded above and below by positive constants and tends to 1. All determinants involved are positive by Lemma 2.2(iv), since $w_k > 0$ for all k and f, h are injective. Lemma 2.3 gives $\log D_n(\sigma) = \log D_n(\sum_k w_k\delta_{Ck^{-p}}) + o(n^2)$. □

Proof of Corollary 1.2. By Lemma 2.4, $\log D_n = n\log m_0 + 2\sum_{j<n}(n-j)\log b_j$, so

$$\frac{2}{n^2}\sum_{j=1}^{n-1}(n-j)\log(j^p b_j) = \frac{1}{n^2}\Big(\log D_n - n\log m_0 + 2p\sum_{j=1}^{n-1}(n-j)\log j\Big).$$

By Theorem 1.1 and Lemma 2.5, the bracket equals $-pn^2\log n + (\kappa_p + \log C)n^2 + pn^2\log n - \frac{3p}{2}n^2 + o(n^2) = \log\big(c(p)C\big)n^2 + o(n^2)$. □

Remark 6.1. Theorem 1.1 gives the error $o(n^2)$ only. For pure powers the proof gives no rate, because the rate of $\mathcal{J}_p(\theta)\to\kappa_p$ in Step 1 of Theorem 5.2 and the factor $A^2/(4p\log M)$ in Theorem 4.3 are not quantified. The numerics of Section 8 are consistent with an error $O(n\log n)$ for pure powers.

7. Hankel determinants of zeta values

Proof of Corollary 1.4. Since $2a+b>1$, every entry $\zeta(a(i+j)+b)$ with $i,j\ge 1$ is a convergent Dirichlet series, and

$$\zeta\big(a(i+j)+b\big) = \sum_{k\ge 1} k^{-(2a+b)}\big(k^{-a}\big)^{(i-1)+(j-1)}.$$

Re-indexing $i,j \mapsto i-1, j-1$, the determinant is $D_n(\sigma)$ for $\sigma = \sum_k k^{-(2a+b)}\delta_{k^{-a}}$. Apply Theorem 1.1 with $p=a$, $f(\xi)=\xi^{-a}$ (so $C=1$ and $\xi^{a+1}f'(\xi)\equiv -a$), and $w_k = k^{-(2a+b)}$, for which $\sum_k w_k < \infty$ and $w_k \geq k^{-s}$ with $s = 2a+b$. The special cases follow from $c(1)=\frac{1}{2}$ and $c(2)=\frac{\pi^2}{16}$. □

For example, $\frac{3a}{2} + \log c(a)$ equals $0.806853\ldots$, $2.516871\ldots$, $4.127795\ldots$, $5.696954\ldots$ for $a=1,2,3,4$ (Table 1).

Remark 7.1 ($a=1$). Monien's $H_n^{(r)}[\zeta] = \det_{1\leq i,j\leq n}\zeta(i+j+r)$ is the case $a=1$, $b=r$ of Corollary 1.4, so $\log H_n^{(r)}[\zeta] = -n^2(\log 2n - \frac{3}{2}) + o(n^2)$ for every real $r > -1$. This is the weak asymptotic form stated in [12] for $r=0,1$.

Remark 7.2 ($a=2$: comparison with closed forms). Baricz and Štampach [1] evaluate Hankel determinants of the Rayleigh functions $\sigma_{2m}(\nu)$. Their Corollary 3 reads "For $n\in\mathbb{N}$, $\nu\notin -\mathbb{N}$, and $\ell\in\{0,1\}$, one has $\det H_n^{(\ell)}(\nu) = 2^{-2n(n+\ell)}\prod_{k=1}^{2n+\ell-1}(\nu+k)^{k-2n-\ell}$", where $H_n^{(\ell)}(\nu)$ has (i,j) entry $\sigma_{2(i+j)+2\ell-2}(\nu)$; their Remark 5 gives $\det H_n^{(2)}(\nu) = 2^{-2n(n+2)}(n+1)(n+\nu+1)\prod_{k=1}^{2n+1}(\nu+k)^{k-2n-2}$; and their (22) reads $\sigma_{2n}(1/2) = \zeta(2n)/\pi^{2n}$. Taking $\nu=\frac{1}{2}$ and rescaling rows and columns by powers of π yields closed forms for $\det_{1\leq i,j\leq n}\zeta(2(i+j)+b)$ with $b = 2\ell-2 \in \{-2,0,2\}$. For example, for $b=0$,

$$\log \det_{1\leq i,j\leq n} \zeta\big(2(i+j)\big) = 2n(n+1)\log\frac{\pi}{2} + \sum_{k=1}^{2n}(k-2n-1)\log\big(k+\tfrac{1}{2}\big).$$

We verified these three closed forms against direct 60-digit determinants for $n\leq 8$ (agreement to at least 10^{-38}; script `checks/bs_check.py`). The asymptotics of the closed form are elementary: with $m=2n+1$, $\sum_{k<m}(m-k)\log(k+\frac{1}{2}) = \frac{m^2}{2}\log m - \frac{3}{4}m^2 + O(m\log m)$ (as in Lemma 2.5), which gives

$$\log \det_{1\leq i,j\leq n} \zeta\big(2(i+j)\big) = -2n^2\log n + \big(3+\log\tfrac{\pi^2}{16}\big)n^2 + O(n\log n),$$

in agreement with Corollary 1.4 for $a=2$, $b=0$, and with a better error term. Numerically, for $\ell=0,1,2$ the quantity $(\log\det + 2n^2\log n)/n^2 - \kappa_2$ equals $-1.33\cdot10^{-4}$, $-3.48\cdot10^{-4}$ and $-5.63\cdot10^{-4}$ at $n=10^5$, and $(n/\log n)$ times it stays between -1.2 and -4.9 for $n\in\{10^2,10^3,10^4,10^5\}$. Thus for $a=2$, $b\in\{-2,0,2\}$, Corollary 1.4 also follows from [1]; the asymptotic statement is not made there.

8. Numerical illustrations

The computations below are floating-point (not certified) and serve only as illustrations. They were produced with the scripts in the accompanying repository and re-run for this draft; the outputs reproduce the stored ones exactly.

Constants and the maximiser. Table 1 lists $c(p)$, u_s and κ_p, together with a Monte Carlo estimate of $\mathcal{L}(\mu^*)$ obtained by sampling μ^* directly (independently of the closed-form value). The density ratio $\mathrm{d}\mu^*/\mathrm{d}\sigma_p$ evaluated by quadrature equals 1 for $x\geq 4c$ to four digits (one exception at $p=0.5$, $x=4000c$, where the quadrature returns 0.9862; the closed form of Theorem 1.3(i) gives exactly 1 there) and is strictly below 1 for $x<4c$.

The averaged law. For $x_k = k^{-p}$ and $w_k = k^{-s}$ truncated to $k\leq K$, the recurrence coefficients b_j were computed in extended precision, and $S_n := \frac{2}{n^2}\sum_{j<n}(n-j)\log(j^p b_j) - \log c(p)$ is shown in Table 2; by Corollary 1.2, $S_n\to 0$. Doubling K changes the tabulated values by at most $5.4\cdot10^{-4}$ ($p=1.5$), $2.8\cdot10^{-5}$ ($p=3$), $1.4\cdot10^{-6}$ ($p=4$) and $7\cdot10^{-8}$ ($p=6$).

p	$c(p)$	$u_s = 2I(p)/\pi$	κ_p	Monte Carlo $\mathcal{L}(\mu^*)$
0.5	0.408248	0.3750	-0.145880	-0.1442 ± 0.0021
1	0.500000	0.5000	0.806853	0.8054 ± 0.0027
1.5	0.566272	0.5798	1.681320	1.6818 ± 0.0034
2	0.616850	0.6366	2.516871	2.5208 ± 0.0042
3	0.689213	0.7132	4.127795	4.1281 ± 0.0058
4	0.738565	0.7628	5.696954	5.6884 ± 0.0075
6	0.801586	0.8235	8.778837	8.7781 ± 0.0109

TABLE 1. Constants of Theorem 1.3. Source: `research/G2/checks/check_energy.out`.

For $p = 0.5$ the atoms decay too slowly for the truncation used (K-doubling changes S_n by 10^{-2}), so that case is not illustrated. For the zeta weights of Corollary 1.4 with $(a, b) = (3, 1)$ and $(4, 1)$ the same statistic is $-0.5544, -0.2181, -0.0750$ and $-0.7452, -0.2922, -0.1004$ at $n = 25, 100, 400$. The identification of $\det \zeta(a(i + j) + b)$ with D_n of $\sum_k k^{-(2a+b)}\delta_{k^{-a}}$ was checked to 12 digits for $(a, b) \in \{(3, 1), (4, 1), (3, 2)\}$ and $n \in \{4, 8, 12\}$.

p	s	K	$n = 25$	50	100	200	400	1600
1.5	1.5	50000	-0.03377	-0.02370	-0.01531	-0.00942	-0.00567	-0.00267
3	3	20000	-0.17266	-0.11244	-0.06962	-0.04161	-0.02423	-0.00782
4	4	6000	-0.27151	-0.17494	-0.10757	-0.06399	-0.03714	–
6	6	2000	-0.46811	-0.29944	-0.18324	-0.10863	–	–

TABLE 2. The averaged statistic S_n. Source: `research/G2/checks/check_averaged.out` and `check_hz.out`.

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